

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
100-ft Ring Road, Bengaluru – 560 085, Karnataka, India

Report on

'QLoRA-Enhanced Domain-Specific LLM Modeling for Hardware Trojan Detection

Submitted by

Debopam Sinha (PES1PG24EC006)

Aug 2025 – May 2026

under the guidance of

Internal Guide

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FACULTY OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM M.TECH

CERTIFICATE

This is to certify that the Report entitled

QLoRA-Enhanced Domain-Specific LLM Modeling for Hardware Trojan Detection

is a bonafide work carried out by

Debopam Sinha (PES1PG24EC006)

In partial fulfillment for the completion of 4rd semester course work in the Program of Study M.Tech in Electronics and Communication Engineering under rules and regulations of PES University, Bengaluru during the period Aug 2025 – May 2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The report has been approved as it satisfies the 4th semester academic requirements in respect of project work.

Signature with date & Seal	*Signature with date & Seal*	*Signature with date & Seal*
Nirmala Devi M	*Dr. Shikha Tripathi*	
Internal Guide	*Chairperson*	*Dean-Faculty of Engg. and Technology*

Name and Signature of the Examiners

1.

2.

DECLARATION

I, **Debopam Sinha,** hereby declare that the report entitled, *QLoRA-Enhanced Domain-Specific LLM Modeling for Hardware Trojan Detection***,** is an original work done by me under the guidance of **Dr. Nirmala Devi M**, Professor, Dept of ECE, and is being submitted in partial fulfillment of the requirements for completion of 4th Semester course work in the Program of Study M.Tech Electronics and Communication Engineering.

PLACE: Bangalore
DATE: 08/05/2026

NAME AND SIGNATURE OF THE CANDIDATE

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Debopam Sinha (PES1PG24EC006)